A. Mother drone — 1 aircraft

Target: 5-inch, 3S, brushless, ArduPilot, GPS/compass, ESP32 video

Item and specification	Qty	Where to buy	Unit price	Total
DolphinRC F405 V3 50A FC + ESC stack	1	SoarSky	$54.59	$54.59
RS2205 2300KV brushless motors, 4-piece variant	1 pack	B-CUBE RC	$20.09	$20.09
5.1-inch Gemfan propellers, 2 CW + 2 CCW	1 set	Pyrodrone — 51499	$3.99	$3.99
BN-880 GPS with integrated compass	1	Exo Express	$12.00	$12.00
AI-Thinker ESP32-CAM + OV2640	1	ElectroPeak	$6.95	$6.95
Powerhobby 3S 1500mAh 45C XT60 LiPo	1	WREKD	$23.99	$23.99
5-inch printed frame, GPS mast and camera mount	1	UCI fabrication	$0	$0
Mother total				$121.6

B. Worker drones — 2 identical aircraft

Each worker: custom ESP32 flight control, 8520 brushed motors, IMU, camera, GNSS, 1S battery

Item and specification	Qty for 2 drones	Where to buy	Unit price	Total
ESP32-WROOM-32D development board	2	ElectroPeak	$4.95	$9.90
GY-521 MPU6050 IMU	2	ElectroPeak	$1.96	$3.92
8520 brushed motors, 4 motors + propellers per kit	2 kits	SpeedyFPV	$4.49	$8.98
AI-Thinker ESP32-CAM OV2640	2	ElectroPeak	$6.95	$13.90
ATGM336H GPS with ceramic antenna	2	Muzi Works	$3.46	$6.92
1S 650mAh 60C PH2.0 LiHV battery	2	WREKD	$8.49	$16.98
MT3608 adjustable 5V boost module	2	ElectroPeak	$0.33	$0.66

Four-channel MOSFET motor-driver components, target allocation	2 sets	DigiKey — SI2302 + UCI prototyping supplies	$2.50	$5.00
Printed frames, prop guards, brackets	2	UCI fabrication	$0	$0
Worker total (both)				$66.2

UCI SENIOR DESIGN PROJECT

Autonomous Search-and-Rescue Drone Swarm

Fall 2026 Project Execution Plan | EECS 159A

University: University of California, Irvine
Project Duration: Fall 2026 – Winter 2027
Current Date: October 7, 2026 (Fall Week 2)
Fall Quarter End: December 11, 2026
Final Project Deadline: March 2027
Team Size: 4 students (2 Computer Engineering, 2 Computer Science & Engineering)
Hardware Budget: $300 maximum
Software Budget: $0 (open-source software)

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1. PROJECT OVERVIEW

1.1 Project Description

The objective of this senior design project is to develop a low-cost, autonomous search-and-rescue drone swarm consisting of three quadcopters and a ground-based artificial intelligence system.

The system will consist of:

- **1 Mother Drone:** A larger quadcopter responsible for GPS navigation, autonomous waypoint missions, and acting as a communication relay.
- **2 Worker Drones:** Smaller quadcopters designed to explore assigned regions, capture images, and transmit sensor information.
- **1 Ground Control Station:** A laptop responsible for artificial intelligence, object detection, mission planning, telemetry monitoring, and swarm coordination.

The system will use ArduPilot, ESP32 microcontrollers, ESP-NOW/Wi-Fi communications, YOLOv8, and a locally hosted large language model.

The primary goal is to demonstrate how multiple inexpensive drones can collaborate to locate simulated victims or objects of interest within a designated search region.

1.2 Project Constraints

1. Maximum hardware budget of $300, including taxes and shipping.
2. All software must be free and open source.
3. All major physical components should be sourced by the end of October where feasible.
4. UCI facilities will be used for fabrication and electronics prototyping.
5. The project must be completed by the end of Winter Quarter 2027.
6. Fall Quarter will focus on design, procurement, early hardware development, and preliminary integration.
7. Winter Quarter will focus on system integration, autonomous behavior, testing, and final demonstrations.
8. All physical flight tests must comply with applicable safety, UCI, and FAA requirements.

1.3 Fall Quarter Objectives

By December 11, 2026, the team should aim to have:

- Finalized and approved system architecture.
- Complete bill of materials and procurement records.
- Mother drone assembled and configured with ArduPilot.
- Mother drone successfully bench-tested and, if authorized, flight-tested.
- At least one functional worker drone prototype.
- Worker drone thrust, mass, power, and flight-control feasibility evaluated.
- ESP32 Wi-Fi and ESP-NOW communications demonstrated.
- Camera stream successfully transmitted to the ground station.
- YOLOv8 object detection running on the ground laptop.
- Three-agent autonomous search simulation running.
- Preliminary integration between hardware and ground software.
- Complete documentation, CAD files, wiring diagrams, and test results.
- A detailed Winter Quarter development plan.

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2. TEAM RESPONSIBILITIES

Team Member 1 — Computer Engineering: Mother Drone Lead

Primary Responsibilities:

- Research and select the ArduPilot-compatible flight controller.

- Select compatible motors, ESCs, propellers, and batteries.
- Design and assemble the mother drone.
- Configure ArduPilot firmware and Mission Planner.
- Install and configure GPS and compass.
- Integrate the physical safety-control receiver.
- Calibrate sensors, motors, and flight modes.
- Develop and execute flight-readiness procedures.
- Perform authorized flight testing.
- Document the mother drone hardware and test results.

Fall Deliverable: Assembled, configured, and tested mother drone.

Team Member 2 — Computer Science & Engineering: Worker Drone Lead

Primary Responsibilities:

- Research micro-drone propulsion and flight-controller options.
- Design worker drone frames and mounting systems.
- Develop the worker hardware mass and power budgets.
- Integrate ESP32, IMU, camera, GPS, and motor drivers.
- Develop embedded sensor and motor-control firmware.
- Implement stabilization and emergency disarm logic.
- Test motors, batteries, regulators, and communications.
- Evaluate custom PCB feasibility.
- Construct and validate at least one worker prototype.

Fall Deliverable: Functional worker prototype with documented flight feasibility.

Team Member 3 — Computer Science & Engineering: AI and Ground Station Lead

Primary Responsibilities:

- Configure the ground laptop environment.
- Install and configure YOLOv8.
- Develop the search-and-rescue object detection pipeline.
- Receive live ESP32 camera streams.
- Develop a ground station dashboard.
- Display drone telemetry and connection status.
- Record detections, timestamps, and confidence scores.
- Integrate camera detection with mission-state messages.
- Measure inference speed, video latency, and detection accuracy.

Fall Deliverable: Operational AI detection and ground monitoring software.

Team Member 4 — Computer Science & Engineering: Swarm Autonomy Lead

Primary Responsibilities:

- Develop the swarm mission coordinator.
- Set up ArduPilot software-in-the-loop (SITL) simulation.
- Implement search-area division and task assignment.
- Design the telemetry and communication protocol.
- Integrate ESP-NOW communication with ground-side software.
- Implement mission state tracking and lost-link handling.
- Develop local LLM-assisted mission planning.
- Simulate multiple autonomous agents.
- Integrate navigation, perception events, and task reassignment.

Fall Deliverable: Working three-agent swarm mission simulation.

Shared Team Responsibilities

All four team members will participate in:

- System architecture decisions.
- Weekly engineering meetings.
- Integration testing.
- Risk management.
- Budget tracking.
- Technical documentation.
- Presentation preparation.
- Demonstration and testing.

Recommended Weekly Commitment: Approximately 6–8 hours per person, adjusted based on course requirements and integration workload.

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3. MASTER PROJECT TIMELINE

WEEK 2: OCTOBER 7–9, 2026

Phase 1 — Project Definition and Research

Primary Goal: Define exactly what the system must accomplish before committing money to hardware.

Tasks

- Conduct a team kickoff meeting.
- Assign primary engineering responsibilities.
- Confirm project expectations with the faculty advisor.
- Clarify whether all three drones must fly simultaneously for the final demonstration.
- Define minimum viable project requirements.
- Write the project problem statement.
- Define measurable engineering success criteria.
- Draw a preliminary overall system block diagram.
- Identify required sensors, processors, motors, and communication interfaces.
- Research verified ArduPilot-compatible flight controllers.
- Compare worker drone propulsion and flight-controller options.
- Estimate worker drone weight, motor thrust, and battery requirements.
- Research battery, GPS, camera, and communication hardware.
- Contact UCI FABWorks and other facilities regarding fabrication availability.
- Investigate loaner flight controllers, radios, and electronics.
- Create a GitHub repository and shared documentation folder.
- Create a hardware budget spreadsheet.
- Install required development software.

Software Setup

- Python
- Visual Studio Code
- Git and GitHub
- ArduPilot SITL
- Mission Planner or a compatible ground control station
- OpenCV
- Ultralytics YOLOv8
- ESP32 development tools

Deliverables by Friday, October 9

1. Written project description.
2. Preliminary architecture diagram.
3. Team responsibility assignments.
4. Initial hardware research and estimated BOM.
5. Development environment established.

6. List of technical risks and unresolved requirements.

WEEK 3: OCTOBER 12–16, 2026

Phase 2 — Final Design and Hardware Selection

Primary Goal: Finalize the aircraft architectures and prepare an approved procurement list.

Mother Drone Tasks

- Choose a verified ArduPilot-compatible FC/ESC stack.
- Select a compatible motor and propeller combination.
- Determine required battery voltage and capacity.
- Choose GPS with compass.
- Select camera and ESP32 interface.
- Confirm frame dimensions and mounting patterns.
- Create wiring and power-distribution diagrams.
- Verify receiver and failsafe requirements.

Worker Drone Tasks

- Complete a preliminary mass budget.
- Calculate estimated thrust-to-weight ratio.
- Compare brushed and brushless motor architectures.
- Choose an IMU and embedded flight controller.
- Select motor drivers and battery regulation.
- Choose camera and GPS modules.
- Determine whether the selected propulsion can carry the entire payload.
- Identify which parts must be prototyped before duplicate purchases.

Software Tasks

- Successfully run ArduPilot SITL.
- Simulate an autonomous waypoint mission.
- Run YOLOv8 on sample search footage.
- Define communication message structure.
- Define unique drone IDs.
- Develop the initial search-area allocation algorithm.

Procurement Tasks

- Build a final itemized BOM.

- Identify vendors and exact part variants.
- Verify component compatibility.
- Obtain final prices including tax and shipping.
- Estimate delivery dates.
- Confirm available UCI equipment.
- Obtain advisor approval before purchasing.

Deliverables by Friday, October 16

1. Final hardware architecture.
2. Approved procurement list.
3. Complete system and electrical block diagrams.
4. Mother drone component compatibility review.
5. Worker drone mass/thrust feasibility analysis.
6. Successful initial software simulation.

MILESTONE 1: DESIGN APPROVAL

No major purchases should occur until component compatibility, worker propulsion feasibility, and the total delivered cost have been reviewed.

WEEK 4: OCTOBER 19–23, 2026

Phase 3 — Hardware Procurement and Mechanical Design

Primary Goal: Order critical components and begin detailed hardware and software development.

Hardware Procurement

- Purchase approved mother flight controller and ESC.
- Purchase mother motors and propellers.
- Purchase GPS/compass module.
- Purchase required ESP32 modules.
- Purchase the first worker propulsion prototype.
- Purchase first worker IMU and power electronics.
- Purchase required batteries from a suitable supplier.
- Obtain or borrow compatible emergency-control equipment.
- Verify and record all order confirmations.
- Record actual costs and estimated delivery dates.

Mechanical Engineering

- Design the mother frame in CAD.
- Design the first worker frame.
- Design camera mounting brackets.
- Design GPS mounting locations.
- Design battery restraints.
- Design electronics compartments.
- Prepare initial 3D printing files.

Electrical Engineering

- Produce mother wiring diagram.
- Produce worker motor-driver circuit.
- Identify required voltage regulators.
- Specify UART, I2C, PWM, and GPIO connections.
- Verify expected motor current and ESC capacity.
- Plan grounding, wiring protection, and battery monitoring.

Software Engineering

- Implement basic ground-station telemetry parsing.
- Implement the simulated search grid.
- Create basic agent assignment logic.
- Create a preliminary detection interface.
- Set up logging for telemetry and mission events.

Deliverables by Friday, October 23

1. Hardware orders placed.
2. Updated budget ledger.
3. Mother and worker CAD designs.
4. Wiring schematics.
5. Working simulated search-area assignment.
6. Initial software repository structure.

MILESTONE 2: PROCUREMENT COMPLETE FOR THE FIRST PROTOTYPES

WEEK 5: OCTOBER 26–30, 2026

Phase 4 — Fabrication and Initial Hardware Testing

Primary Goal: Begin physical construction and determine whether the worker drone is mechanically feasible.

Mother Drone Tasks

- Print first frame prototype.
- Inspect dimensions and motor mounting points.
- Verify structural integrity.
- Test flight controller power input.
- Check ESC connectivity.
- Verify GPS and compass connections.
- Prepare assembly layout.

Worker Drone Tasks

- Print initial micro-drone frame.
- Weigh all electronics.
- Measure total projected flying mass.
- Build a safe motor-thrust test fixture.
- Measure motor thrust and current draw.
- Evaluate expected hover throttle and battery load.
- Check motor and propeller clearance.
- Decide whether the propulsion configuration is viable.

Software Tasks

- Demonstrate first ESP32 camera stream if parts have arrived.
- Run YOLOv8 on the stream or substitute footage.
- Simulate three drones operating in separate search regions.
- Display basic mission progress.
- Log telemetry messages and timestamps.

Deliverables by Friday, October 30

1. Printed initial aircraft frames.
2. Worker thrust test results.
3. Updated worker mass budget.
4. Initial hardware electrical tests.
5. Functional AI image-processing pipeline.
6. Three-drone search simulation.

MILESTONE 3: WORKER PROPULSION FEASIBILITY GATE

If the proposed worker motors cannot safely lift the required hardware with adequate control margin, redesign the worker propulsion system before buying parts for Worker 2.

WEEK 6: NOVEMBER 2–6, 2026

Phase 5 — Electronics Assembly and Embedded Systems Integration

Primary Goal: Assemble the mother drone and worker electronics while establishing basic communication.

Mother Drone Tasks

- Install motors and ESC.
- Mount the flight controller.
- Install GPS/compass.
- Install camera and ESP32 module.
- Install radio receiver.
- Complete wiring.
- Verify voltage levels and polarity.
- Connect the FC to ground-control software.
- Verify sensor readings.
- Verify motor identification and direction with propellers removed.

Worker Drone Tasks

- Assemble worker motor-driver circuitry.
- Connect IMU to ESP32.
- Read and calibrate IMU measurements.
- Connect and test motor control outputs.
- Configure battery voltage monitoring.
- Verify regulated power to electronics.
- Connect GPS receiver.
- Implement basic motor mixing.
- Implement motor-disarm behavior.
- Test battery-current and motor-temperature limits.

Communications Tasks

- Establish ESP-NOW connection between development boards.
- Transmit unique vehicle identifiers.
- Transmit heartbeat packets.
- Transmit sensor telemetry.
- Add message timestamps.
- Measure packet delivery.
- Detect communication loss.
- Define a safe response to link failure.

Software Tasks

- Display telemetry in the ground station.
- Run YOLOv8 on available camera feeds.
- Build a basic mission-state interface.
- Simulate battery warnings and vehicle disconnection.

Deliverables by Friday, November 6

1. Assembled mother drone electronics.
2. Worker motor and IMU bench prototype.
3. Verified ESP-NOW communication.
4. Working telemetry interface.
5. Electrical and safety test records.

MILESTONE 4: ELECTRICAL INTEGRATION COMPLETE

WEEK 7: NOVEMBER 9–13, 2026

Phase 6 — Initial Flight Validation and System Testing

Primary Goal: Begin controlled flight validation where approved and integrate real hardware telemetry.

Mother Drone Tasks

- Complete sensor calibration.
- Verify battery and voltage monitoring.
- Configure flight modes.
- Test radio disarm and failsafe functions.
- Complete flight-readiness inspection.
- Obtain required testing authorization.
- Perform initial supervised hover testing if ready.
- Record flight logs and video.
- Identify stability, vibration, and power issues.

Worker Drone Tasks

- Verify IMU axis orientation.
- Verify motor mixing.
- Test stabilization outputs.
- Evaluate flight-control loop timing.

- Validate the loss-of-link disarm function.
- Conduct staged testing only when propulsion and safety requirements are met.
- Record motor-response and stability measurements.

Ground Station Tasks

- Receive mother drone telemetry.
- Display location and operating state.
- Record camera stream frame rate.
- Evaluate camera transmission latency.
- Demonstrate YOLOv8 detections.
- Log detection events.

Swarm Tasks

- Simulate three vehicles executing separate search routes.
- Demonstrate mission reassignment.
- Test disconnected vehicle handling.
- Integrate simulated mission states with ground-station telemetry.

Deliverables by Friday, November 13

1. Mother drone initial flight results or documented blocker.
2. Worker stabilization test results.
3. Ground station displaying real telemetry.
4. Camera transmission performance measurements.
5. Integrated simulation demonstration.

MILESTONE 5: INITIAL SYSTEM VALIDATION

WEEK 8: NOVEMBER 16–20, 2026

Phase 7 — Prototype Refinement and Design Freeze

Primary Goal: Resolve major hardware problems and lock the architecture for Winter integration.

Mother Drone Tasks

- Fix issues discovered during initial testing.
- Refine ArduPilot parameters.
- Verify GPS position estimates.

- Validate emergency-control procedures.
- Test waypoint mission behavior in simulation.
- Conduct additional authorized flight tests if appropriate.

Worker Drone Tasks

- Improve stabilization and motor responsiveness.
- Reduce unnecessary airframe weight.
- Improve electronic mounting.
- Resolve power instability.
- Evaluate whether a custom PCB would improve the system.
- Finalize the worker design.
- Approve second worker purchase only after prototype validation.

Software Tasks

- Integrate live detections into the mission coordinator.
- Add mission event logging.
- Implement task reassignment when an agent becomes unavailable.
- Test simulated collision avoidance and no-fly boundaries.
- Develop a more complete ground station interface.

Documentation Tasks

- Update complete CAD designs.
- Update wiring schematics.
- Record firmware and software versions.
- Update BOM with actual costs.
- Update risk assessment.
- Conduct advisor review.

Deliverables by Friday, November 20

1. Validated mother design.
2. Finalized worker architecture.
3. Updated software architecture.
4. Approved second worker build strategy.
5. Updated test evidence and risk register.

MILESTONE 6: HARDWARE ARCHITECTURE FREEZE

After this deadline, avoid major design changes unless required for safety or fundamental functionality.

WEEK 9: NOVEMBER 23–27, 2026

Phase 8 — System Demonstration and Report Development

Primary Goal: Produce a repeatable prototype demonstration and draft the Fall technical report.

Thanksgiving holidays: November 26–27. Complete important team testing before the holiday.

Hardware Tasks

- Repair outstanding mother drone issues.
- Verify repeatable system startup.
- Check all electrical connections.
- Verify camera and telemetry operation.
- Continue Worker 1 testing.
- Assemble Worker 2 if validated parts are available.
- Document the current physical configuration of each aircraft.

Software Tasks

- Run YOLOv8 detection using an actual ESP32 camera feed.
- Demonstrate telemetry transmission to the laptop.
- Run three-agent simulated search.
- Demonstrate automatic region assignment.
- Demonstrate simulated lost-agent reassignment.
- Record end-to-end event logs.

Documentation Tasks

- Draft technical report.
- Update project architecture diagrams.
- Collect photographs of hardware.
- Export CAD files.
- Include electrical schematics.
- Include BOM and actual expenses.
- Summarize completed tests.
- Identify remaining Winter tasks.
- Begin presentation slides.

Deliverables by Wednesday, November 25

1. Repeatable preliminary demonstration.
2. Draft Fall technical report.

3. Current hardware and software documentation.
4. Preliminary presentation.
5. List of unfinished requirements.

MILESTONE 7: DEMONSTRATION FREEZE

WEEK 10: NOVEMBER 30 – DECEMBER 4, 2026

Phase 9 — Final Verification and Fall Project Completion

Primary Goal: Complete engineering work and prepare the final Fall submission.

Testing Tasks

- Perform final mother hardware bench checks.
- Confirm stable telemetry and camera streaming.
- Re-run approved flight tests where appropriate.
- Verify worker sensor and motor functions.
- Validate emergency shutdown behavior.
- Run final integrated ground station demonstration.
- Demonstrate three-agent mission allocation in simulation.
- Record successful and unsuccessful test results.
- Measure performance metrics.

Documentation Tasks

- Finalize technical report.
- Finalize bill of materials.
- Finalize CAD drawings.
- Finalize wiring diagrams.
- Complete design justification.
- Complete risk analysis.
- Summarize engineering calculations.
- Document testing procedures and outcomes.
- Finalize Winter Quarter execution plan.

Presentation Tasks

- Prepare final presentation slides.
- Assign speaking sections.
- Record demonstration footage.
- Rehearse the full presentation.

- Obtain advisor feedback.
- Make final corrections.

Deliverables by Friday, December 4

1. Completed Fall engineering prototype package.
2. Final draft technical report.
3. Finished demonstration materials.
4. Completed presentation.
5. Detailed Winter project plan.

MILESTONE 8: FALL ENGINEERING WORK COMPLETE

FINALS WEEK: DECEMBER 5–11, 2026

Phase 10 — Submission and Winter Handoff

Primary Goal: Submit all required materials and prepare the team to resume in Winter.

Tasks

- Submit required EECS 159A assignments by their actual Canvas deadlines.
- Deliver project presentation and demonstration.
- Submit final technical report.
- Back up GitHub repository and project files.
- Archive hardware firmware versions.
- Archive CAD and PCB files.
- Archive test data and flight logs.
- Save purchase receipts and budget records.
- Document remaining problems.
- Assign Winter Week 1 tasks.
- Properly disconnect and store LiPo batteries.
- Confirm that all teammates have access to project files.

Deliverables by December 11

1. Complete Fall report and presentation.
2. Functional preliminary prototype and recorded test evidence.
3. Updated project repository.
4. Complete engineering documentation.
5. Winter Quarter development roadmap.

MILESTONE 9: FALL QUARTER COMPLETE

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4. HARDWARE PROCUREMENT AND BUDGET MANAGEMENT

4.1 Budget Distribution

Maximum Total Hardware Budget: $300

Category	Planning Budget
Mother drone	$121.61
Worker drones (2 total)	$66.26
Radio, programming, and shared hardware	$70.86
Remaining taxes, shipping, and contingency	$41.27
Total	**$300.00**

The $258.73 component subtotal is a preliminary sourcing estimate, not a verified final checkout amount. All taxes, delivery charges, and replacement components must remain within the $300 limit.

4.2 Purchasing Schedule

October 7–16

- Complete hardware selection.
- Confirm component compatibility.
- Check UCI inventory and equipment availability.
- Compare vendors.
- Obtain final delivered-price estimates.
- Obtain design approval.

October 19–23

- Purchase critical mother drone components.
- Purchase one worker prototype's essential components.
- Order communication electronics and required sensors.
- Place battery orders after compatibility checks.

October 26–30

- Receive and inspect early components.
- Complete initial propulsion testing.
- Resolve any component or motor incompatibility.

November 2–20

- Purchase verified replacement parts if needed.
- Purchase Worker 2 components only after validating Worker 1.
- Reserve sufficient funds for tax, shipping, and unforeseen costs.

4.3 UCI Resources

The team should attempt to source the following through UCI facilities:

- 3D-printed airframes.
- Motor mounts.
- Camera mounts.
- GPS mounts.
- Propeller guards.
- Landing gear.
- Electronics enclosures.
- Battery holders.
- Prototype PCB materials.
- Soldering stations.
- Solder, flux, wire, and heat shrink.
- Multimeters.
- Oscilloscopes.
- Bench power supplies.
- LiPo balance chargers and safety equipment.
- General machining and fabrication tools.

Relevant campus and external resources:

- UCI FABWorks: https://manufacturing.uci.edu/index.php/fabworks-2/
- UCI IDMI Machine Shop: https://manufacturing.uci.edu/index.php/facilities/machineshop/
- UCI EECS Capstone: https://sites.uci.edu/eecscapstone/
- JLCPCB: https://jlcpcb.com/
- PCBWay: https://www.pcbway.com/

- McMaster-Carr: https://www.mcmaster.com/

Free access or materials should be confirmed with the facility; do not assume all services are automatically covered.

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5. SOFTWARE DEVELOPMENT PLAN

5.1 Software Architecture

Ground Laptop

- YOLOv8 for object detection.
- OpenCV for image processing.
- Python for telemetry and mission logic.
- Local LLM for high-level mission assistance.
- Ground station dashboard.
- Mission coordination system.
- Search-area allocation algorithm.

Mother Drone

- ArduPilot flight-control firmware.
- GPS and compass.
- ESP32 camera interface.
- Telemetry connection.
- Receiver-based safety control.

Worker Drones

- ESP32 embedded firmware.
- IMU sensor processing.
- Motor control and stabilization.
- GPS telemetry.
- ESP32 camera streaming.
- Communication loss monitoring.
- Emergency disarm system.

5.2 Software Implementation Order

1. Establish development environment.

2. Set up repository and version control.
3. Run ArduPilot simulation.
4. Implement simulated waypoint navigation.
5. Install and test YOLOv8.
6. Implement image detection on prerecorded footage.
7. Develop simulated search-area allocation.
8. Define drone telemetry messages.
9. Develop ESP32 Wi-Fi communications.
10. Develop ESP-NOW telemetry and control messages.
11. Implement camera streaming.
12. Integrate live telemetry into the laptop.
13. Build the ground station dashboard.
14. Integrate detection events with mission coordinator.
15. Implement mission reassignment.
16. Simulate connection failures and low-battery behavior.
17. Conduct end-to-end hardware/software integration.
18. Record benchmarks and test results.

The local LLM should provide high-level assistance, not unrestricted real-time motor control. Each aircraft requires deterministic stabilization and safety behavior.

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6. HARDWARE AND SYSTEM TESTING REQUIREMENTS

6.1 Mother Drone Tests

- Verify electrical polarity.
- Verify safe power distribution.
- Verify flight-controller startup.
- Verify ESC connectivity.
- Verify motor order and direction.
- Calibrate sensors.
- Validate GPS and compass.
- Verify receiver connectivity.
- Verify emergency disarm.
- Verify communication-loss failsafe.
- Verify battery monitoring.
- Complete flight-readiness checklist.
- Perform authorized supervised hover testing.

- Analyze logs and document stability.

6.2 Worker Drone Tests

- Verify electronics mass.
- Verify motor thrust.
- Verify battery current and voltage.
- Validate regulator output.
- Validate IMU data.
- Verify motor mixing.
- Test control-loop timing.
- Test emergency disarm.
- Verify camera stream.
- Verify GPS telemetry.
- Verify ESP-NOW connection.
- Evaluate flight stabilization.
- Validate safe failure behavior.

6.3 Ground Station Tests

- Receive telemetry from a real ESP32.
- Display live drone status.
- Receive camera frames.
- Record video frame rate and latency.
- Run YOLOv8 detection.
- Detect designated mock rescue targets.
- Log object-detection confidence.
- Track detection timestamps.
- Simulate three assigned search regions.
- Demonstrate mission reassignment after loss of one simulated agent.
- Record results and error conditions.

6.4 Preliminary Performance Targets

Metric	Fall Target
Camera test duration	60 seconds continuous streaming
ESP-NOW test	100 consecutive messages with packet-loss measurement
AI test dataset	At least 20 labeled test images
Swarm simulation	Three active simulated agents

Search regions	Three distinct assigned areas
Failure test	Detect lost simulated agent and reassign its task
Mother flight	Supervised controlled hover if flight-ready and authorized
Worker flight	Demonstrated stabilization or documented redesign decision
Total hardware expenditure	$300 maximum delivered

These are internal engineering targets and may be adjusted after baseline measurements.

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7. RISK MANAGEMENT

Risk	Impact	Mitigation
Worker motors cannot lift payload	Critical	Measure actual thrust before duplicate purchases
Flight controller incompatible with ArduPilot	High	Verify exact board and firmware target before ordering
ESP32 camera adds excessive weight	High	Optimize payload and frame before flight
Battery insufficient for propulsion	High	Validate current, voltage sag, and expected flight time
Worker stabilization development takes too long	High	Establish early technical gate and simulation fallback
Wireless interference	Medium–High	Test throughput, channel configuration, and packet loss
Autonomous navigation instability	High	Start with SITL and supervised individual flight
Hardware shipping delays	High	Order critical components early
Budget overrun	High	Compare total delivered prices before purchase

Component failure	Medium	Preserve funds and test components individually
Campus fabrication unavailable	Medium	Confirm authorization and scheduling early
Unsafe simultaneous flight	Critical	Require individual flight validation and authorization

Contingency Strategy

If all three drones cannot achieve reliable autonomous flight within the budget, the team should preserve a technically meaningful final demonstration consisting of:

1. One operational ArduPilot mother drone.
2. Two physical worker prototypes with verified sensors, communications, and documented flight-control progress.
3. A functioning laptop-based AI vision system.
4. A three-agent autonomous swarm simulation.
5. A working interface between actual hardware telemetry and the simulated mission coordinator.

This fallback should be confirmed with the faculty advisor rather than assumed to satisfy all capstone requirements.

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8. FALL QUARTER TECHNICAL DOCUMENTATION

The team should maintain the following throughout the quarter.

Required Engineering Documents

1. Project proposal and problem statement.
2. Functional and nonfunctional requirements.
3. System architecture diagram.
4. Hardware block diagram.
5. Communications and data-flow diagram.
6. Mother drone wiring diagram.
7. Worker drone wiring diagram.

8. CAD models and manufacturing drawings.
9. Itemized bill of materials.
10. Budget and purchasing records.
11. Power and mass budgets.
12. Motor thrust evaluation.
13. Software architecture documentation.
14. Communication protocol specification.
15. Flight-readiness and safety procedures.
16. Subsystem test procedures.
17. Test results and analysis.
18. Risk register and mitigation plan.
19. Design decisions and revision history.
20. Winter Quarter implementation schedule.

Suggested GitHub Repository Structure

```
drone-swarm/
├── docs/
├── hardware/
│   ├── mother/
│   ├── worker/
│   └── pcb/
├── firmware/
│   ├── mother-bridge/
│   └── worker/
├── ground-station/
│   ├── vision/
│   ├── telemetry/
│   └── dashboard/
├── swarm/
│   ├── planning/
│   ├── agents/
│   └── safety/
├── simulation/
└── tests/
```

All design files, software, diagrams, test logs, and documentation should be stored in the shared repository or linked project folder.

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9. WEEKLY PROJECT MANAGEMENT PROCESS

Weekly Team Meeting

Recommended Frequency: Once per week, approximately 30 minutes.

Meeting Agenda

1. Review last week's goals.
2. Confirm completed tasks.
3. Identify incomplete or blocked tasks.
4. Review hardware and software progress.
5. Review budget spending.
6. Review safety and technical risks.
7. Assign next week's tasks.
8. Confirm milestone deadlines.

Weekly Integration Session

Recommended Frequency: Once per week, approximately two hours.

During this session, members should combine their subsystem work, test interfaces, and identify problems that do not appear during isolated development.

Weekly Status Report Format

Week:
Date:
Completed Tasks:
Tasks In Progress:
Blocked Tasks:
Hardware Purchases:
Budget Remaining:
Testing Completed:
Technical Risks:
Next Week's Priorities:
Assigned Owners:

Every task should have one primary owner and a deadline.

10. WINTER QUARTER PREVIEW

Fall Quarter is not the final project deadline. The purpose of the Fall plan is to create a sufficiently mature system so Winter can focus on performance and integration.

January 4–15, 2027

Goal: Hardware Completion

- Finalize unresolved hardware.
- Assemble remaining drone components.
- Validate independent electronics and flight controls.
- Verify communications.
- Repair mechanical and electrical failures.

January 18–29, 2027

Goal: System Integration

- Integrate all drone telemetry with the ground station.
- Integrate camera video feeds.
- Connect navigation information to the mission coordinator.
- Test target detection and event handling.
- Continue individually authorized flight tests.

February 1–12, 2027

Goal: Swarm Intelligence

- Finalize search-area allocation.
- Implement mission reassignment.
- Integrate the local LLM for high-level task planning.
- Test agent coordination.
- Simulate failures and recovery behavior.

February 15–26, 2027

Goal: Coordinated System Testing

- Verify stable operation of individual aircraft.
- Test multi-agent communications.
- Test search-task coordination.
- Perform supervised cooperative tests only when approved.
- Measure detection performance and communication reliability.
- Refine mission behavior.

March 1–12, 2027

Goal: Final Demonstration and Validation

- Complete repeatable demonstration.
- Evaluate system reliability.
- Record test results.
- Record demonstration videos.
- Complete final technical documentation.
- Prepare final presentation.
- Correct critical remaining issues.

March 13–19, 2027

Goal: Final Submission

- Submit required final capstone deliverables.
- Deliver final presentation.
- Demonstrate the system.
- Archive source code and documentation.
- Complete the project handoff.

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11. MAJOR PROJECT MILESTONES

Date	Milestone	Expected Result
October 9	Project Definition	Scope, roles, architecture, initial research
October 16	Design Approval	Approved BOM, compatibility, feasibility
October 23	Prototype Procurement	Critical initial hardware ordered
October 30	Worker Feasibility	Motor thrust and weight evaluation

November 6	Electronics Integration	Working sensors, motor control, telemetry
November 13	Initial Validation	Mother flight/bench results, worker stabilization
November 20	Architecture Freeze	Verified hardware design and Winter direction
November 25	Demonstration Freeze	Repeatable preliminary system demonstration
December 4	Engineering Completion	Final report draft, testing, presentation
December 11	Fall Quarter Complete	Formal submission and Winter handoff
January 22	Winter Hardware Checkpoint	All required prototypes assembled and tested
February 26	System Integration Target	Coordinated functionality and test evidence
March 12	Final Demonstration Target	Working system and final report ready
March 19	Winter Quarter End	Capstone complete, subject to actual course deadlines

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12. IMMEDIATE ACTION PLAN — OCTOBER 7–9

The following should be completed within the next two days.

- Schedule the first full team meeting.
- Assign the four engineering roles.
- Confirm the faculty advisor and Fall course deliverables.
- Write the final project description.
- Establish the minimum viable final demonstration.
- Create GitHub repository and shared folder.
- Create the first system architecture diagram.
- Verify mother flight-controller compatibility.
- Complete the initial worker propulsion weight estimate.
- Contact UCI fabrication facilities.
- Investigate university equipment loans.

- Create the $300 budget tracker.
- Install the software development environment.
- Run the first simulated ArduPilot mission.
- Schedule the October 16 hardware design review.

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13. FALL QUARTER DEFINITION OF SUCCESS

By December 11, 2026, the team should have completed the fundamental engineering design and have evidence that its major subsystems are operational.

Minimum Successful Fall Outcome

- One assembled ArduPilot mother drone with verified electrical and navigation systems.
- One worker drone prototype with tested propulsion, sensors, electronics, and communications.
- A working ESP32 camera interface.
- A ground laptop running AI detection.
- A simulated three-drone autonomous search mission.
- Verified communication between physical hardware and ground software.
- Complete technical documentation.
- A clear, achievable Winter Quarter roadmap.

Stretch Goals

- Successful supervised mother drone autonomous waypoint flight.
- Successful stabilized flight of Worker 1.
- Completed assembly of Worker 2.
- Simultaneous telemetry from multiple physical drone prototypes.
- Real camera detections triggering mission-state updates.
- Early coordinated behavior between physical and simulated agents.

The team should prioritize reliable engineering, documented verification, and a functioning integrated prototype over attempting unvalidated simultaneous autonomous flight too early.

Final Objective: Enter Winter Quarter with the aircraft architecture established, expensive engineering uncertainties resolved, and the main AI and mission-coordination components already operating. This gives the team the best opportunity to complete a credible autonomous search-and-rescue drone swarm by March 2027 within the $300 hardware limit.